

EN3160: Image Processing and Machine Vision

Assignment 1: Intensity Transformations and Neighborhood Filtering

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GitHub Repository: github.com/RumethRanhinda/EN3160-Vision-UoM

1. Intensity Transformation Implementation

```
[2]: import cv2
import numpy as np
import matplotlib.pyplot as plt

def intensity_transform(im, breakpoints):
    bp = np.array(breakpoints)
    xp = bp[:, 0] # Input intensities
    yp = bp[:, 1] # Output intensities

    transformed_im = np.interp(im.flatten(), xp, yp).reshape(im.shape) # Apply interpolation

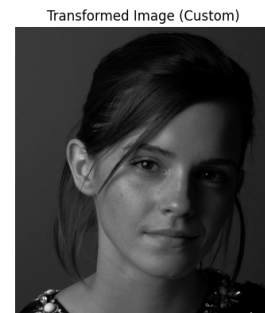
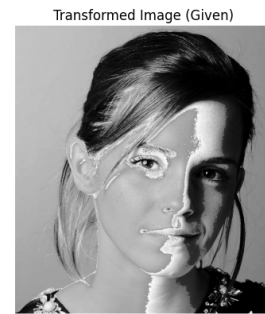
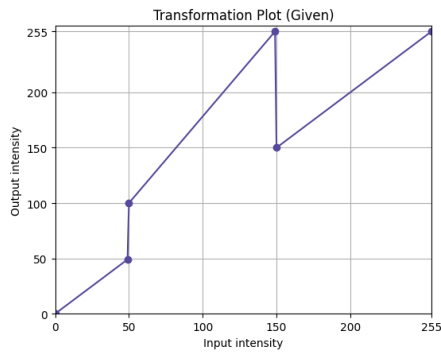
    return transformed_im.astype(np.uint8)

img = cv2.imread('Images/emma.jpeg', cv2.IMREAD_GRAYSCALE)
bp_given = [[0, 0], [49, 49], [50, 100], [149, 255], [150, 150], [255, 255]] # Provided ↴
# ↵ breakpoints
bp_custom = [[0, 0], [50, 20], [175, 100], [255, 255]] # Custom breakpoints

transformed_given = intensity_transform(img, bp_given)
transformed_custom = intensity_transform(img, bp_custom)

grid_ticks = [0, 50, 100, 150, 200, 255] # Grid ticks for the plots
fig, axes = plt.subplots(2, 3, figsize=(15, 9))
# Given Breakpoints
bp_given_arr = np.array(bp_given)
axes[0, 0].plot(bp_given_arr[:,0], bp_given_arr[:,1], marker='o', color='#55499C')
axes[0, 0].set_title("Transformation Plot (Given)")
axes[0, 0].set_xlabel("Input intensity"); axes[0, 0].set_ylabel("Output intensity")
axes[0, 0].set_xlim([0, 255]); axes[0, 0].set_ylim([0, 260])
# Apply custom grid
axes[0, 0].set_xticks(grid_ticks); axes[0, 0].set_yticks(grid_ticks); axes[0, 0].grid(True)
axes[0, 1].imshow(img, cmap='gray', vmin=0, vmax=255)
axes[0, 1].set_title("Original Image"); axes[0, 1].axis('off')
axes[0, 2].imshow(transformed_given, cmap='gray', vmin=0, vmax=255)
axes[0, 2].set_title("Transformed Image (Given)"); axes[0, 2].axis('off')
```

```
# Custom Breakpoints
bp_custom_arr = np.array(bp_custom)
axes[1, 0].plot(bp_custom_arr[:,0], bp_custom_arr[:,1], marker='o', color='#55499C')
axes[1, 0].set_title("Transformation Plot (Custom)")
axes[1, 0].set_xlabel("Input intensity"); axes[1, 0].set_ylabel("Output intensity")
axes[1, 0].set_xlim([0, 255]); axes[1, 0].set_ylim([0, 260])
axes[1, 0].set_xticks(grid_ticks); axes[1, 0].set_yticks(grid_ticks); axes[1, 0].grid(True)
axes[1, 1].imshow(img, cmap='gray', vmin=0, vmax=255)
axes[1, 1].set_title("Original Image"); axes[1, 1].axis('off')
axes[1, 2].imshow(transformed_custom, cmap='gray', vmin=0, vmax=255)
axes[1, 2].set_title("Transformed Image (Custom)"); axes[1, 2].axis('off')
plt.tight_layout(); plt.show()
```



The given intensity transformation made high intensities above 150 darker in a discontinuous way which explains the artifact seen in the output image.

The custom transformation was implemented with the intention of highlighting the freckles within the bright area of the face by providing a wider range for brighter pixels.

2. Accentuate White and Gray Matter

The idea is to first find the threshold value of intensity which determines white and dark matter, and in each case providing a wider range via intensity transformation, for input pixel values within that threshold range.

- **For White Matter:** 150 - 255
- **For Dark Matter:** 128 - 190

```
[3]: brain_img = cv2.imread('Images/brain.jpeg', cv2.IMREAD_GRAYSCALE)

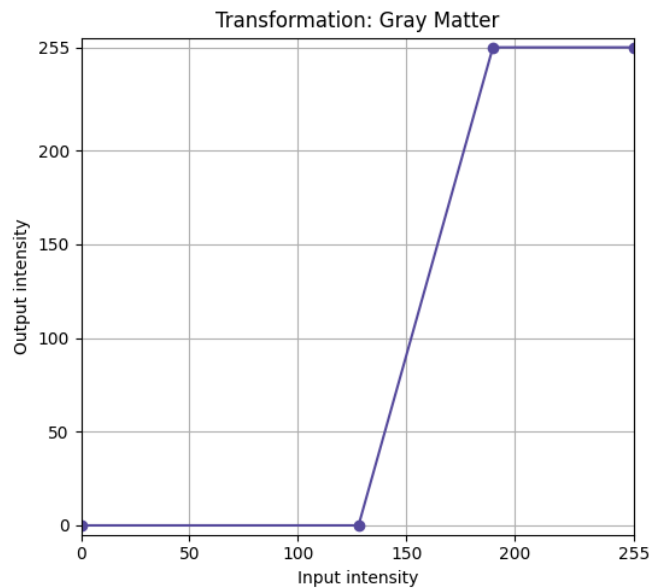
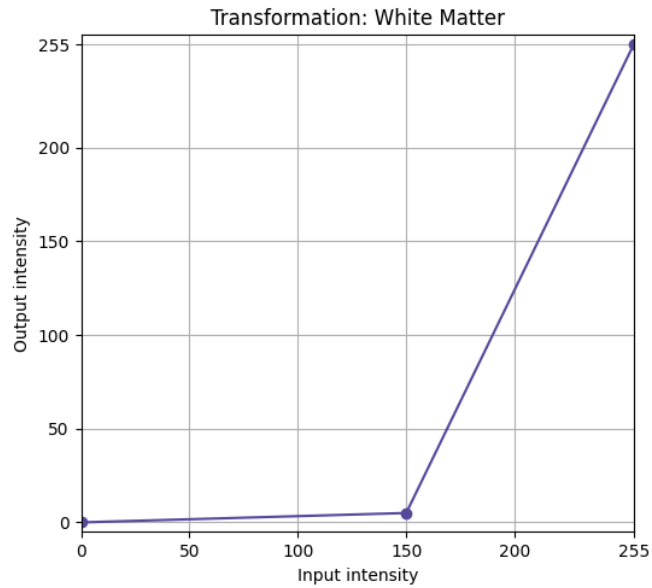
bp_white_matter = [[0, 0], [150, 5], [255, 255]]
bp_gray_matter = [[0, 0], [128, 0], [190, 255], [255, 255]]

wm_accentuuated = intensity_transform(brain_img, bp_white_matter)
gm_accentuuated = intensity_transform(brain_img, bp_gray_matter)

grid_ticks = [0, 50, 100, 150, 200, 255]
fig, axes = plt.subplots(2, 2, figsize=(10, 10))

# White Matter Transformation
bp_wm_arr = np.array(bp_white_matter)
axes[0, 0].plot(bp_wm_arr[:,0], bp_wm_arr[:,1], marker='o', color='#55499C')
axes[0, 0].set_title("Transformation: White Matter")
axes[0, 0].set_xlabel("Input intensity"); axes[0, 0].set_ylabel("Output intensity")
axes[0, 0].set_xlim([0, 255]); axes[0, 0].set_ylim([-5, 260])
axes[0, 0].set_xticks(grid_ticks); axes[0, 0].set_yticks(grid_ticks); axes[0, 0].grid(True)
axes[0, 1].imshow(wm_accentuuated, cmap='gray', vmin=0, vmax=255)
axes[0, 1].set_title("White Matter Accentuated"); axes[0, 1].axis('off')

# Gray Matter Transformation
bp_gm_arr = np.array(bp_gray_matter)
axes[1, 0].plot(bp_gm_arr[:,0], bp_gm_arr[:,1], marker='o', color='#55499C')
axes[1, 0].set_title("Transformation: Gray Matter"); axes[1, 0].set_xlabel("Input intensity")
axes[1, 0].set_ylabel("Output intensity"); axes[1, 0].set_xlim([0, 255]); axes[1, 0].
    ↪set_ylim([-5, 260])
axes[1, 0].set_xticks(grid_ticks); axes[1, 0].set_yticks(grid_ticks); axes[1, 0].grid(True)
axes[1, 1].imshow(gm_accentuuated, cmap='gray', vmin=0, vmax=255)
axes[1, 1].set_title("Gray Matter Accentuated"); axes[1, 1].axis('off')
plt.tight_layout(); plt.show()
```



3. Gamma Correction

```
[4]: img_bgr = cv2.imread('Images/legs.jpg')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB) # For display in Matplotlib

img_lab = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2LAB) # Convert to L*a*b* color space
L, a, b = cv2.split(img_lab)

gamma = 0.5

# Normalize L to [0, 1], apply power law, and scale back to [0, 255]
L_normalized = L / 255.0
L_corrected = np.power(L_normalized, gamma)
L_corrected = np.uint8(L_corrected * 255)

# Merge the corrected L plane with the original a and b planes
```

```

img_lab_corrected = cv2.merge((L_corrected, a, b))
img_corrected_rgb = cv2.cvtColor(img_lab_corrected, cv2.COLOR_LAB2RGB)

# Compute histograms
hist_L_orig = cv2.calcHist([L], [0], [None], [256], [0, 256])
hist_L_corr = cv2.calcHist([L_corrected], [0], [None], [256], [0, 256])

fig, axes = plt.subplots(2, 2, figsize=(12, 10))

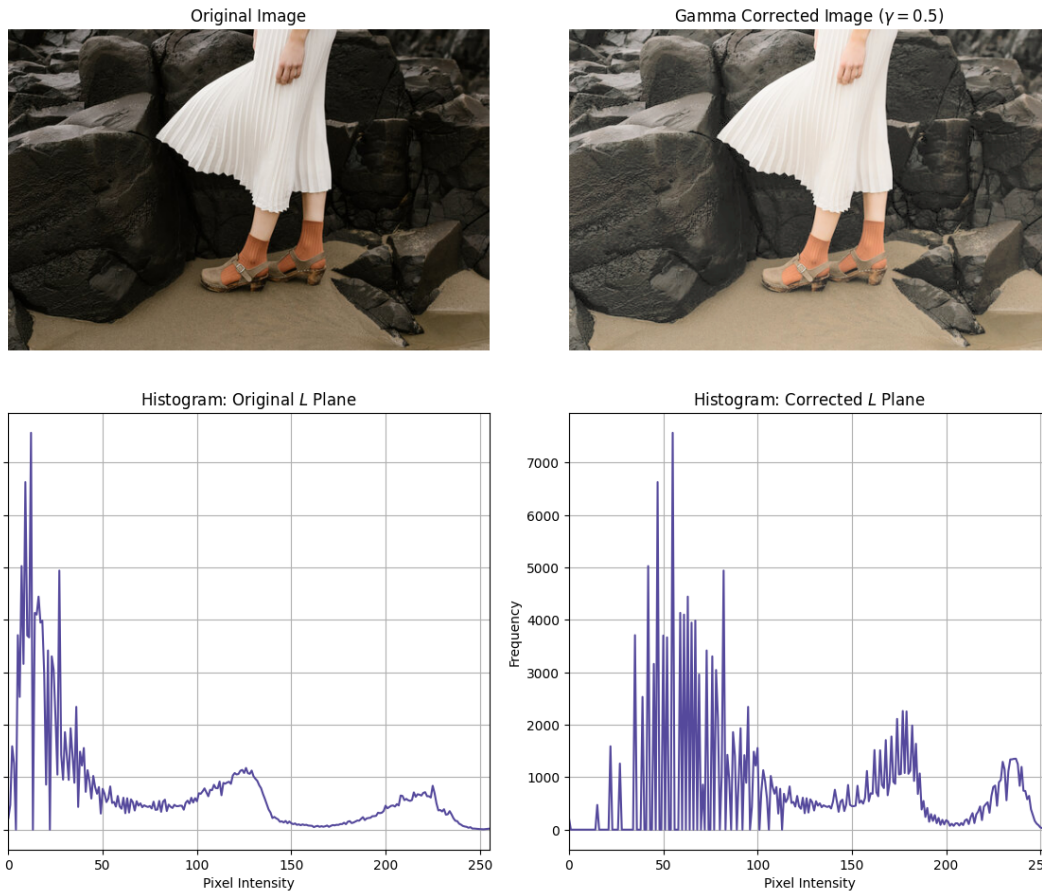
# Original Image
axes[0, 0].imshow(img_rgb); axes[0, 0].set_title("Original Image"); axes[0, 0].axis('off')

# Gamma Corrected Image
axes[0, 1].imshow(img_corrected_rgb); axes[0, 1].set_title(rf"Gamma Corrected Image_
↳ ($\gamma={gamma}$)")
axes[0, 1].axis('off')

# Histogram of Original L Plane
axes[1, 0].plot(hist_L_orig, color='#55499C')
axes[1, 0].set_title("Histogram: Original $L$ Plane")
axes[1, 0].set_xlabel("Pixel Intensity"); axes[1, 0].set_ylabel("Frequency")
axes[1, 0].set_xlim([0, 255]); axes[1, 0].grid(True)

# Histogram of Corrected L Plane
axes[1, 1].plot(hist_L_corr, color='#55499C')
axes[1, 1].set_title("Histogram: Corrected $L$ Plane")
axes[1, 1].set_xlabel("Pixel Intensity"); axes[1, 1].set_ylabel("Frequency")
axes[1, 1].set_xlim([0, 255]); axes[1, 1].grid(True)
plt.tight_layout(); plt.show()

```



4. Enhancing Vibrance using Intensity Transformation

```
[5]: img_bgr = cv2.imread('Images/spider.jpeg')

if img_bgr is not None:
    img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

    # Convert to HSV color space
    img_hsv = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2HSV)
    H, S, V = cv2.split(img_hsv)

    a = 0.55
    sigma = 70.0

    # Convert S plane to float32 to prevent overflow during mathematical operations
    S_float = S.astype(np.float32)

    gaussian_term = np.exp(-((S_float - 128.0)**2) / (2.0 * sigma**2))
    S_transformed = S_float + a * 128.0 * gaussian_term
    S_transformed = np.clip(S_transformed, 0, 255).astype(np.uint8)

    # Recombine the three planes
    img_hsv_enhanced = cv2.merge((H, S_transformed, V))
    img_enhanced_rgb = cv2.cvtColor(img_hsv_enhanced, cv2.COLOR_HSV2RGB)
```

```

# Generate the 1D intensity transformation curve
x_vals = np.arange(0, 256, dtype=np.float32)
y_vals = x_vals + a * 128.0 * np.exp(-((x_vals - 128.0)**2) / (2.0 * sigma**2))
y_vals = np.clip(y_vals, 0, 255)

fig, axes = plt.subplots(1, 3, figsize=(15, 5))

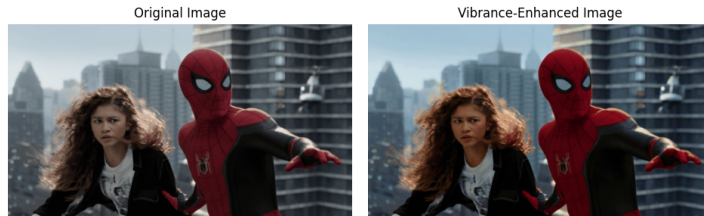
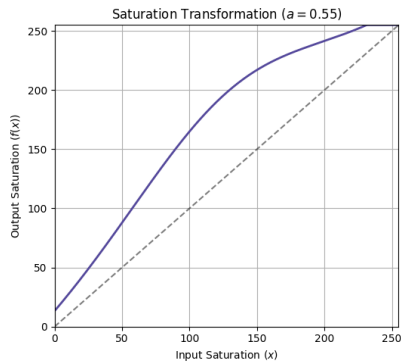
# Plot 1: Transformation Curve
axes[0].plot(x_vals, y_vals, color='#55499C', linewidth=2)
axes[0].plot([0, 255], [0, 255], 'k--', alpha=0.5) # Reference linear line
axes[0].set_title(rf"Saturation Transformation ($a={a}$)")
axes[0].set_xlabel("Input Saturation ($x$)")
axes[0].set_ylabel("Output Saturation ($f(x)$)")
axes[0].set_xlim([0, 255]); axes[0].set_ylim([0, 255])
axes[0].grid(True)

# Plot 2: Original Image
axes[1].imshow(img_rgb)
axes[1].set_title("Original Image")
axes[1].axis('off')

# Plot 3: Enhanced Image
axes[2].imshow(img_enhanced_rgb)
axes[2].set_title("Vibrance-Enhanced Image")
axes[2].axis('off')

plt.tight_layout()
plt.show()
else:
    print("Image not found")

```



Increasing a improves the vibrancy of the image. Ranging a from 0.5 to 0.8 did not result in significant changes. Although above 0.8 the image starts to become unrealistically colourful. To avoid major clipping of saturation values at 255, a is chosen to be 0.55.

5. Custom Histogram Equalization

```

[6]: def histogram_equalization(img):
      # Compute the histogram
      hist, bins = np.histogram(img.flatten(), 256, [0, 256])

```

```

cdf = hist.cumsum()

# 3. Mask out zero values in the CDF to ignore empty bins during normalization
cdf_masked = np.ma.masked_equal(cdf, 0)

# 4. Normalize the CDF to the range [0, 255]
cdf_normalized = (cdf_masked - cdf_masked.min()) * 255 / (cdf_masked.max() - cdf_masked.
↪min())

# 5. Fill the masked values back with 0 and cast to uint8
cdf_final = np.ma.filled(cdf_normalized, 0).astype(np.uint8)

# 6. Map the original pixel intensities to the new equalized values
img_eq = cdf_final[img]

return img_eq, hist, cdf_final

img = cv2.imread('Images/shells.tif', cv2.IMREAD_GRAYSCALE)

if img is not None:
    img_equalized, hist_orig, mapping_function = histogram_equalization(img)

    # Compute the histogram
    hist_eq, _ = np.histogram(img_equalized.flatten(), 256, [0, 256])
    fig, axes = plt.subplots(2, 2, figsize=(12, 10))

    # Original Image
    axes[0, 0].imshow(img, cmap='gray', vmin=0, vmax=255)
    axes[0, 0].set_title("Original Image")
    axes[0, 0].axis('off')

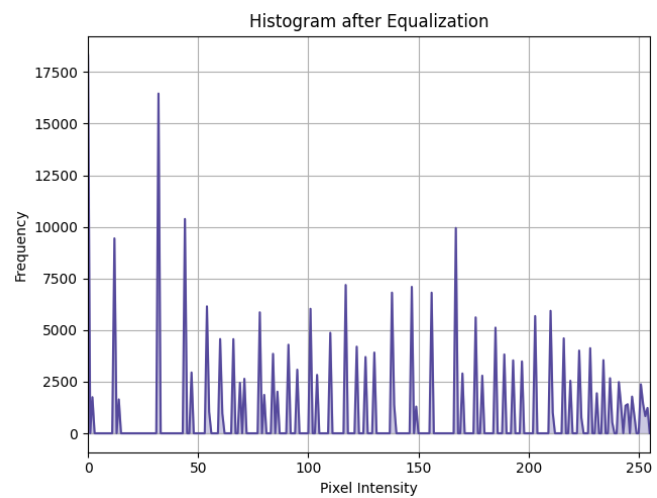
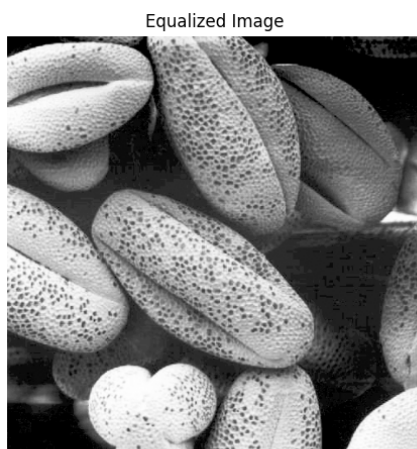
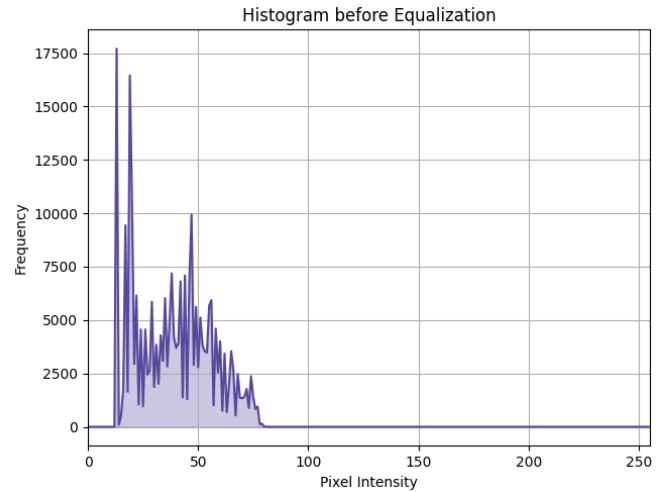
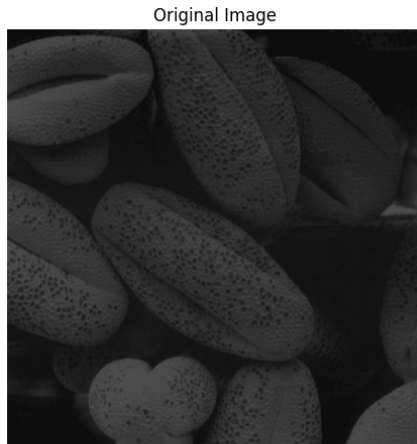
    # Original Histogram
    axes[0, 1].plot(hist_orig, color='#55499C')
    axes[0, 1].fill_between(range(256), hist_orig, color='#55499C', alpha=0.3)
    axes[0, 1].set_title("Histogram before Equalization")
    axes[0, 1].set_xlabel("Pixel Intensity")
    axes[0, 1].set_ylabel("Frequency")
    axes[0, 1].set_xlim([0, 255])
    axes[0, 1].grid(True)

    # Equalized Image
    axes[1, 0].imshow(img_equalized, cmap='gray', vmin=0, vmax=255)
    axes[1, 0].set_title("Equalized Image")
    axes[1, 0].axis('off')

    # Equalized Histogram
    axes[1, 1].plot(hist_eq, color='#55499C')
    axes[1, 1].fill_between(range(256), hist_eq, color='#55499C', alpha=0.3)
    axes[1, 1].set_title("Histogram after Equalization")
    axes[1, 1].set_xlabel("Pixel Intensity")
    axes[1, 1].set_ylabel("Frequency")
    axes[1, 1].set_xlim([0, 255])
    axes[1, 1].grid(True)

    plt.tight_layout()
    plt.show()
else:
    print("Image not found")

```

6. Neighborhood Filtering

```
[7]: img_bgr = cv2.imread('Images/jennifer.jpeg')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

# convert to HSV color space and split into H, S, V planes
img_hsv = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2HSV)
H, S, V = cv2.split(img_hsv)

# Threshold mask on saturation plane / Otsu's method calculates the threshold value.
_, mask = cv2.threshold(S, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)

# Obtain the foreground
fg_v = cv2.bitwise_and(V, V, mask=mask)

# Calculate the histogram of the Value plane, strictly using the mask to ignore the background
hist_fg = cv2.calcHist([V], [0], mask, [256], [0, 256]).flatten()

# (d) Obtain the cumulative sum of the histogram using np.cumsum
```

```

cdf = np.cumsum(hist_fg)

# (e) Histogram-equalize the foreground[cite: 1]
# Mask out 0s to prevent empty bins from skewing the normalization
cdf_m = np.ma.masked_equal(cdf, 0)
cdf_normalized = (cdf_m - cdf_m.min()) * 255 / (cdf_m.max() - cdf_m.min())
cdf_final = np.ma.filled(cdf_normalized, 0).astype(np.uint8)

# Create a copy of the V plane and map the equalized values ONLY to the foreground coordinates
V_eq = V.copy()
fg_coords = mask > 0
V_eq[fg_coords] = cdf_final[V[fg_coords]]

# (f) Extract the background and add with the histogram equalized foreground[cite: 1]
mask_inv = cv2.bitwise_not(mask)
bg_v = cv2.bitwise_and(V, V, mask=mask_inv)
fg_eq_v = cv2.bitwise_and(V_eq, V_eq, mask=mask)

# Add the untouched background and the equalized foreground back together
final_v = cv2.add(bg_v, fg_eq_v)

# Recombine the H, S, and new V planes, and convert back to RGB for display
final_hsv = cv2.merge((H, S, final_v))
final_rgb = cv2.cvtColor(final_hsv, cv2.COLOR_HSV2RGB)

# Display the required planes and outputs[cite: 1]
fig, axes = plt.subplots(2, 3, figsize=(15, 10))

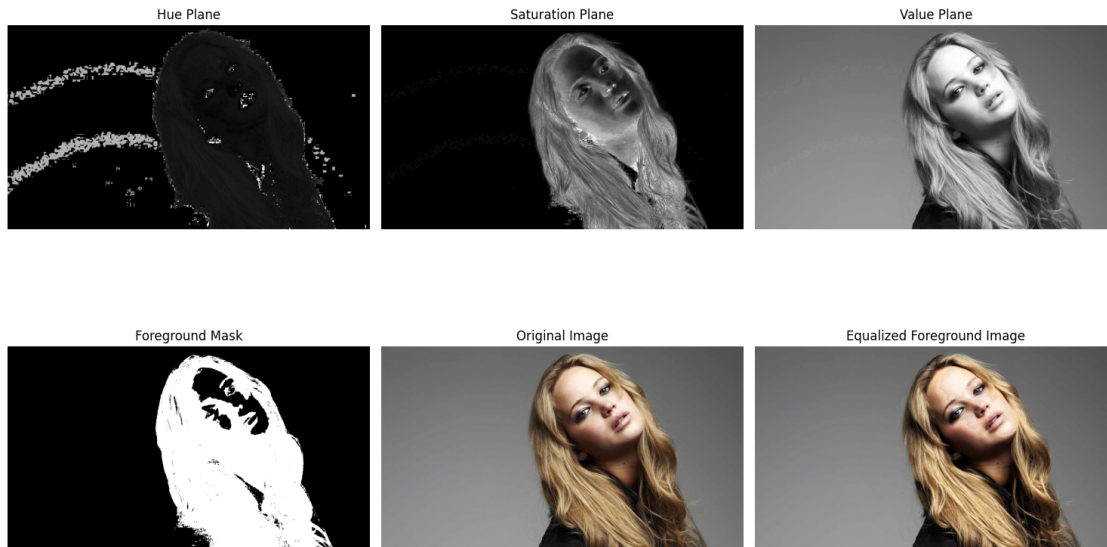
# Row 1: The individual HSV planes in grayscale
axes[0, 0].imshow(H, cmap='gray'); axes[0, 0].set_title('Hue Plane')
axes[0, 1].imshow(S, cmap='gray'); axes[0, 1].set_title('Saturation Plane')
axes[0, 2].imshow(V, cmap='gray'); axes[0, 2].set_title('Value Plane')

# Row 2: Mask, Original, and Final Result
axes[1, 0].imshow(mask, cmap='gray'); axes[1, 0].set_title('Foreground Mask')
axes[1, 1].imshow(img_rgb); axes[1, 1].set_title('Original Image')
axes[1, 2].imshow(final_rgb); axes[1, 2].set_title('Equalized Foreground Image')

for ax in axes.flatten():
    ax.axis('off')

plt.tight_layout()
plt.show()

```



The saturation plane is able to be used for the foreground mask because the background is dull and thus have lower saturation. The thresholding do miss a portion of the black dress the subject is wearing, which can be ignored.

7. Sobel Filtering

```
[ ]: def custom_filter2d(image, kernel):
    """
    Custom 2D spatial filter
    """
    i_h, i_w = image.shape
    k_h, k_w = kernel.shape
    pad_h, pad_w = k_h // 2, k_w // 2

    # Pad the image with zeros to handle edge pixels
    padded_img = np.pad(image, ((pad_h, pad_h), (pad_w, pad_w)), mode='constant')
    output = np.zeros_like(image, dtype=np.float32)

    # Correlation
    for i in range(k_h):
        for j in range(k_w):
            output += kernel[i, j] * padded_img[i:i+i_h, j:j+i_w]

    return output

img = cv2.imread('Images/einstein.jpeg', cv2.IMREAD_GRAYSCALE)

# Convert image to float32 to prevent overflow and preserve negative gradients
img_f = img.astype(np.float32)

# Define the standard 3x3 Sobel X kernel
sobel_x = np.array([[ 1,  0, -1],
                    [ 2,  0, -2],
                    [ 1,  0, -1]], dtype=np.float32)
```

```

# Using cv2.filter2D
filtered_a = cv2.filter2D(img_f, cv2.CV_32F, sobel_x)

# Using the custom 2D filter
filtered_b = custom_filter2d(img_f, sobel_x)

# Using the property of separability
kernel_col = np.array([[1], [2], [1]], dtype=np.float32)
kernel_row = np.array([[1, 0, -1]], dtype=np.float32)

# Apply sequentially: filter with the column, then filter that result with the row
temp_c = custom_filter2d(img_f, kernel_col)
filtered_c = custom_filter2d(temp_c, kernel_row)

# Helper function to format floating-point gradients for visual display
def format_for_display(gradient_img):
    abs_grad = np.abs(gradient_img)
    return np.clip(abs_grad, 0, 255).astype(np.uint8)

# Plot all results
fig, axes = plt.subplots(1, 4, figsize=(20, 5))

axes[0].imshow(img, cmap='gray')
axes[0].set_title('Original Image')

axes[1].imshow(format_for_display(filtered_a), cmap='gray')
axes[1].set_title('(a) cv2.filter2D')

axes[2].imshow(format_for_display(filtered_b), cmap='gray')
axes[2].set_title('(b) Custom filter2D')

axes[3].imshow(format_for_display(filtered_c), cmap='gray')
axes[3].set_title('(c) Separable Filtering')

for ax in axes:
    ax.axis('off')

plt.tight_layout(); plt.show()

```



Mathematical Note on Separability: The spatial Sobel filter is computationally expensive as the kernel size increases. However, the 3×3 Sobel X matrix is separable, meaning it can be decomposed into the outer product of a column vector and a row vector:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} * [1 \quad 0 \quad -1]$$

By passing a 1×3 filter and then a 3×1 filter sequentially over the image, the number of multiplication operations per pixel drops from K^2 (9 operations) to $2K$ (6 operations), which significantly optimizes performance in embedded vision applications while producing the exact same mathematical gradient output.